VIPPS · CASE INTERVIEW

# Does VIP create customer loyalty?

Assessing VIP's impact on spending, retention, and engagement.

| 419 | 350,567 | 30 | 269 |
|---|---|---|---|
| Users in the dataset | Purchases | Merchants | Experiment users |

01 · DEFINE

## How do we measure loyalty?

Define "Keep coming back" in a way that reflects how customers use the service.

02 · EVALUATE

## Does VIP change behaviour?

Compare spending, retention, and engagement across VIP and non VIP users, experiment & control groups.

03 · RECOMMEND

## What should happen next?

Turn the findings into a decision, follow-up analysis, and success measures.

- From defining loyalty to evaluating impact and recommending action

Repository ↗

EXECUTIVE SUMMARY

# Invites drive signups, not retention.

VIP enrollment is associated with higher activity, but the invitation experiment does not show a significant retention lift.
The clearest next step is to test targeted cross-selling and improve measurement.

01 · DEFINE SUCCESS

## Keep coming back

We define repeat engagement through three metrics:

- **Survival:** The user made a purchase in the last **28** days.
- **Cadence:** The user maintains the expected purchase frequency.
- **Spend:** The user continues to make purchases and generate GMV.

Together, these metrics capture whether users remain active, return regularly, and continue spending on the platform.

02 · EARLY VIP ADOPTERS

## Customers who adopted VIP were already more active

Future VIP customers spent **1.09×** more per week before launch. After enrollment, purchases rose **8.4%** and GMV **5.5%** within user, driven by in-person activity.

03 · INVITATION EXPERIMENT

## Experiment boosts signup, not retention

Signup increased from **8.0%** to **68.1%** **(+60.2 pp)** (**$p < 0.001$**). After 16 weeks, ITT effects were **4.7%** on purchases and **-0.5%** on GMV; no retention metric was significant.

04 · WHAT TO DO NEXT

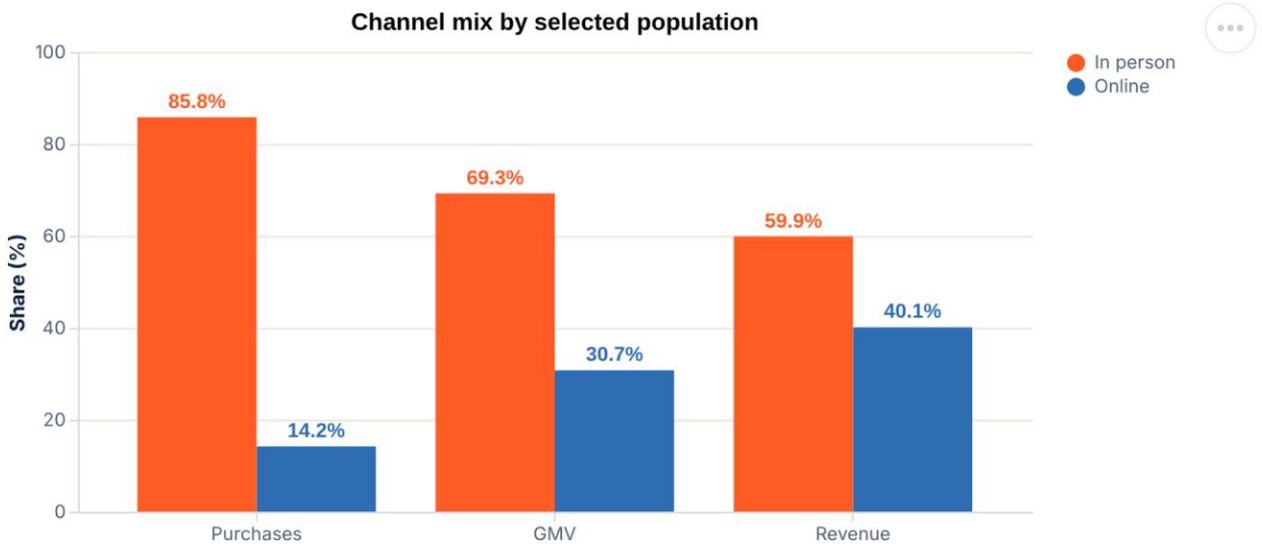
## Test cross-sell and anti-churn

Online take rate is **1.95%** versus **1.29%** for in-person. Shifting customers to online could increase revenue. To reduce churn, try 14-day reminders instead of batch invitations.

**BOTTOM LINE** Invitations are effective at converting users into VIPs; retention impact remains unproven.

CHANNEL ECONOMICS

# Two distinct products in one: daily habit and high-value retail




CHANNEL COMPARISON

## Two distinct spending patterns

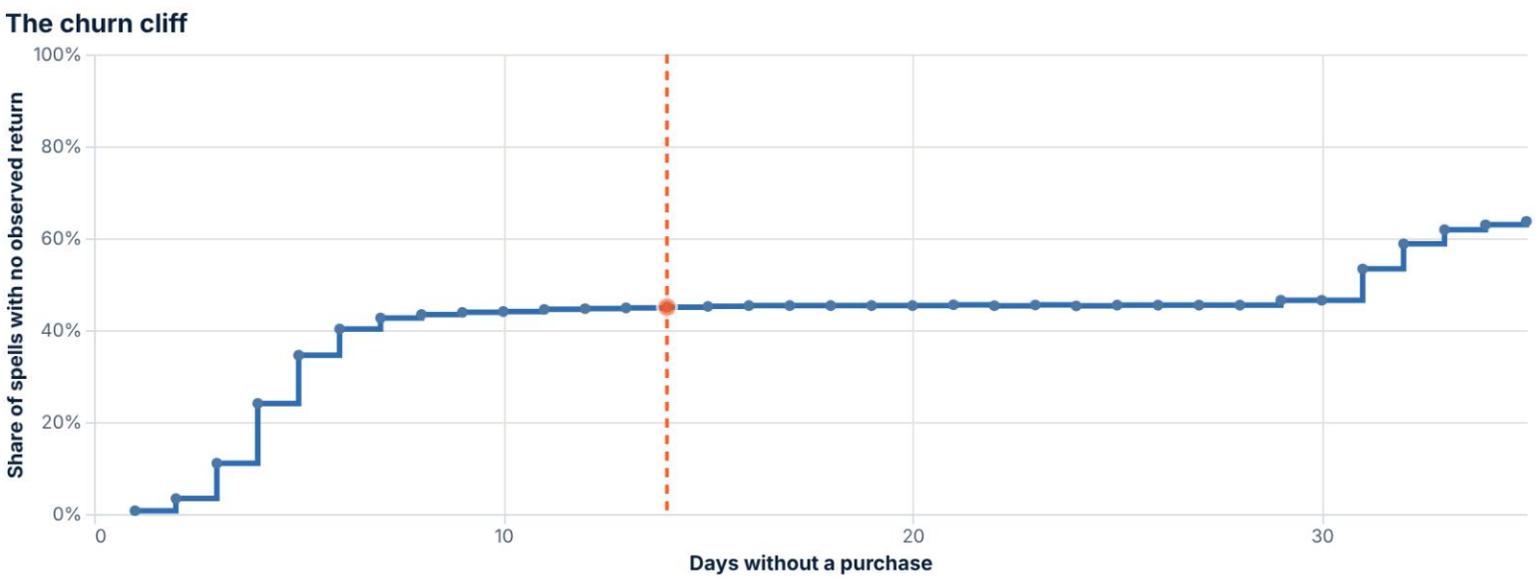
| Metric | In person | Online |
|---|---|---|
| Active users | 384 | 385 |
| Mean ticket | 375 NOK | 1,007 NOK |
| Median ticket | 241 NOK | 395 NOK |
| Take rate | 1.29% | **1.95%** |

Online represents **14.2%** of purchases, but **30.7%** of GMV and **40.1%** of service revenue. Its take rate is **1.95%**, compared with **1.29%** in person.

DEFINING RETENTION

# What does it mean to keep coming back?

## The churn cliff




No observed return by 14 days

**45%**

Across 271 purchase spells

Users active within the 28-day window

**299**

Last purchase within 28 days of year-end

RETENTION FRAMEWORK

## Three layers of “keep coming back”

01 / SURVIVAL

**Recent activity**

Purchased within the selected survival window.

02 / CADENCE

**Expected frequency**

Weekly in-person and monthly online activity.

03 / SPEND

**Customer value**

Purchase frequency and GMV.

Based on the RFM framework

JAN–APR · BEFORE VIP LAUNCH

# Future VIPs were already more engaged

Before VIP launched, users who later enrolled were already spending more and purchasing more often than users who never enrolled.

AVERAGE WEEKLY GMV · JAN–APR · BEFORE VIP LAUNCH

**1.09×**

Future VIP average vs. Never-VIP

N=189 vs. N=196 · Welch t-test p = 0.191




AVERAGE WEEKLY PURCHASES · JAN–APR · BEFORE VIP LAUNCH

**1.10×**

Future VIP average vs. Never-VIP

N=189 vs. N=196 · Welch t-test p = 0.084



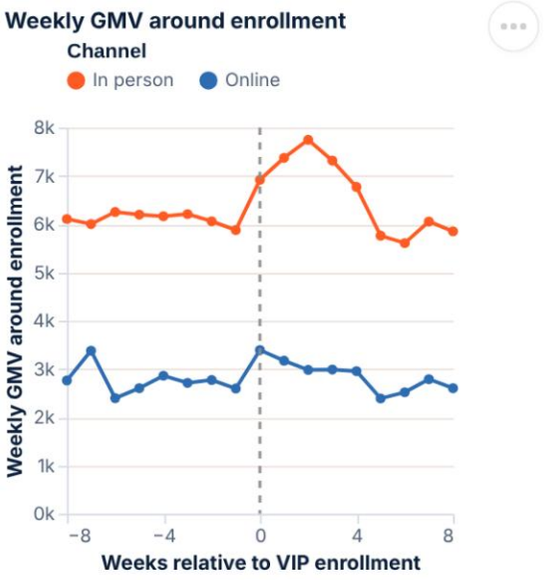

**How to interpret this:** This pre-existing difference is evidence of selection bias: the full-year gap cannot be interpreted as the effect of VIP membership alone.

Q1 · DO PEOPLE WHO JOIN VIP BUY MORE?

# VIP enrollment coincides with higher in-person activity

Activity for the selected cohort before and after enrollment. The dashed vertical line marks the enrollment week.




BEFORE VS. AFTER ENROLLMENT

| Metric | Before | After | Change | p-value |
|---|---|---|---|---|
| **Purchases / week** | 19.1 | 20.7 | **+8.4%** | 0.000 |
| **In-person GMV / week** | 6,108 NOK | 6,560 NOK | **+7.4%** | 0.000 |
| **Online GMV / week** | 2,758 NOK | 2,798 NOK | **+1.4%** | 0.752 |
| **Total GMV / week** | 8,866 NOK | 9,358 NOK | **+5.5%** | 0.014 |

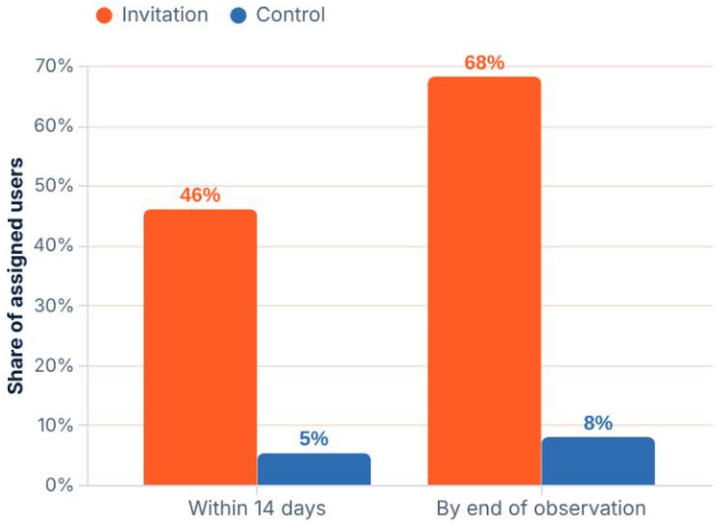
**How to read this (N=177):** Each user's average weekly activity before enrollment is compared with their average after enrollment. The change is an association, not proof that VIP caused it, because users choose when to join.

Q2 · DOES INVITING PEOPLE TO JOIN VIP HELP IN MAKING THEM COME BACK?

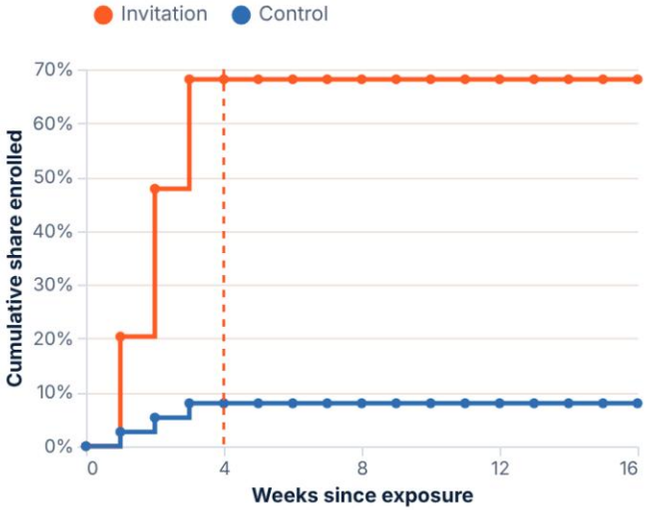
# The invitation increased VIP enrollment

Compare enrollment between assigned experiment arms, then follow how the gap develops over time.

### Enrollment rate by experiment arm




### Cumulative enrollment over time




BASELINE CHARACTERISTICS

| Baseline | Invitation | Control | p-value |
|---|---|---|---|
| **June transactions (selection variable)** | 81.65 | 81.79 | 0.980 |
| **Pre-exposure purchases / week** | 18.14 | 18.05 | 0.940 |

Invited and Control arms had near-identical June purchase frequency prior to assignment (p = 0.903), confirming balanced randomization.

CUMULATIVE ENROLLMENT · WEEK 4

**68.1% invitation** vs. **8.0% control**

**Difference: +60.2 percentage points**

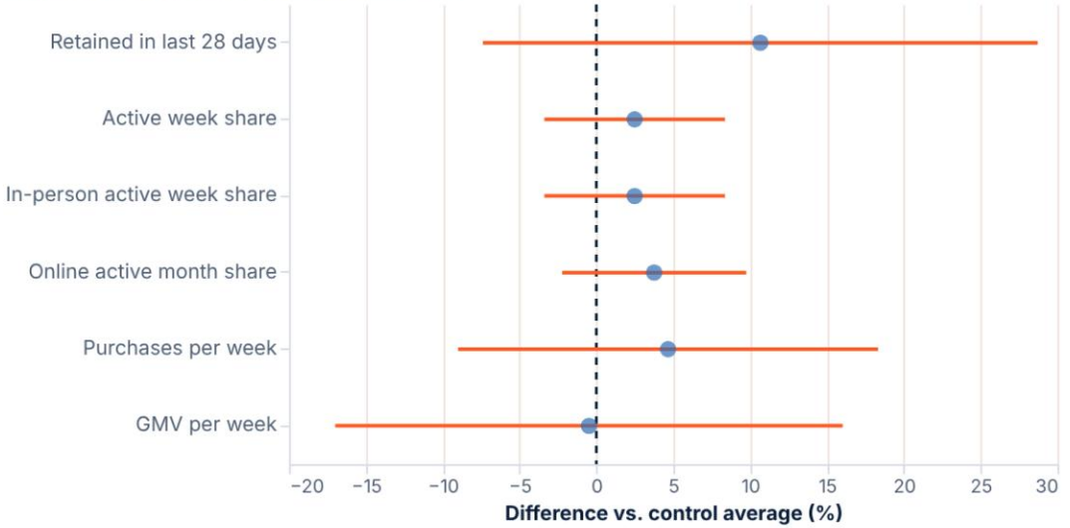
Ever-enrollment difference across the observation period: +60.2 pp (p < 0.001).

Q2 · DOES INVITING PEOPLE TO JOIN VIP HELP IN MAKING THEM COME BACK?

# The invitation did not show a clear change in retention or spend

Compare post-exposure activity and spending between invitation and control users.

**Invitation vs. control · 16-week window**




OUTCOME ESTIMATES

| Metric | Control | Invitation | Diff. [95% CI] | p |
|---|---|---|---|---|
| **Retained in last 28 days** | 66.4% | 73.5% | +10.7% [-7.4, +28.7] | 0.248 |
| **Active week share** | 88.9% | 91.1% | +2.5% [-3.4, +8.4] | 0.407 |
| **In-person active week share** | 88.9% | 91.1% | +2.5% [-3.4, +8.4] | 0.407 |
| **Online active month share** | 88.3% | 91.6% | +3.8% [-2.2, +9.7] | 0.220 |
| **Purchases per week** | 16.7 | 17.5 | +4.7% [-9.0, +18.3] | 0.504 |
| **GMV per week** | 7,945 NOK | 7,907 NOK | -0.5% [-17.0, +16.0] | 0.955 |

INTERPRETATION

**All 95% confidence intervals include zero, so we cannot conclude the invitation changed retention or spending.**

Points show estimated differences; lines show 95% confidence intervals. June-frequency groups use pre-experiment activity.

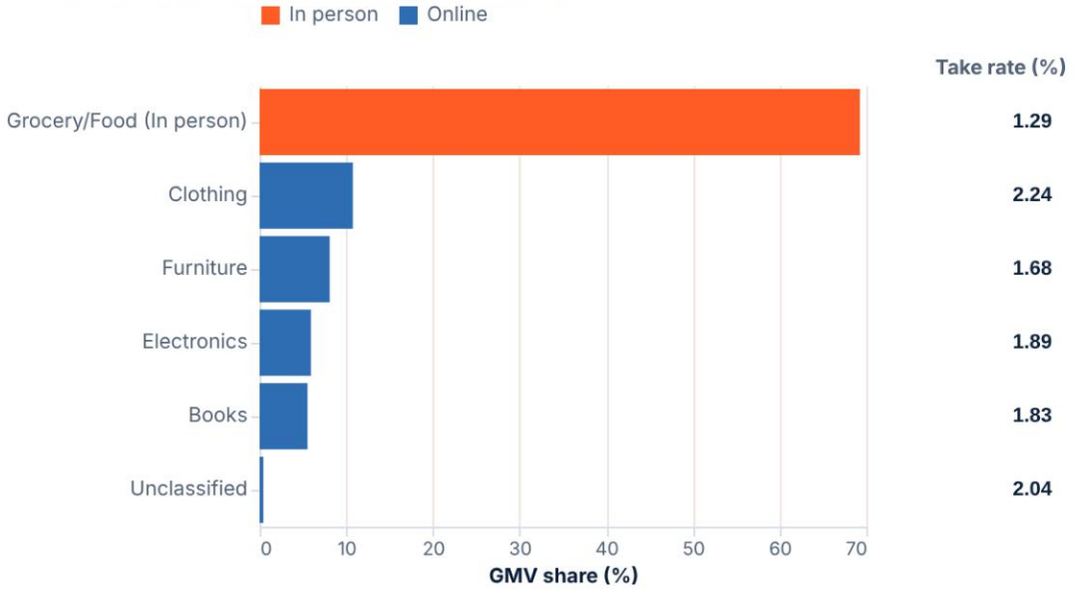
Q3 · RECOMMENDATIONS & NEXT STEPS

# Turn VIP membership into an active habit

Sector GMV and take rates can help identify where online cross-sell may be worth testing.

## GMV share by sector

Bar length shows GMV share; the right column shows take rate




GMV share describes transaction value mix; take rate is shown as a label and is not encoded by bar length.

### IDEAS TO TEST

1. Send a reminder after a period of inactivity.
2. Test point multipliers to encourage online purchases.
3. Track invite opens, clicks, purchases, and point redemption.

### ILLUSTRATIVE ONLINE CROSS-SELL SCENARIO

What if each VIP made **1.0 extra online purchases per month?**

| 1 · Extra purchases / year | 2 · Incremental online GMV | 3 · Estimated fee revenue | 4 · Point cost |
|---|---|---|---|
| **2,340** | **NOK 2,359,799** | **NOK 46,025** | **– NOK 11,799** |
| VIPs × 12 × 1.0 | Purchases × 1,008 NOK average ticket | GMV × 1.95% take rate | GMV × 50 bps assumed cost |

ESTIMATED FEE REVENUE LESS POINT COST

Fee revenue – point cost

**NOK 34,226**

Illustrative only. Assumes 195 VIP users (2025 total). Take rate is average of all online take rates.

Q3 · RECOMMENDATIONS & NEXT STEPS

# What we left for later

Focus the analysis on reliable, decision-ready evidence. Revisit these areas when the data can support stronger conclusions.

## DEFERRED ANALYSES

01 · CARD-LEVEL ANALYSIS

### cards.csv

**Why deferred**
Card status is only known at the 2026-01-01 cutoff, so it would leak future information into 2025 behaviour. It is also card-level, not user-level.

**Revisit when:** card history (issued, blocked, cancelled) exists.

02 · AGE SEGMENTATION

### 16 age buckets

**Why deferred**
Splitting 419 users into 16 four-year bands leaves about 26 per band, and only ~7 per experiment arm — too few to compare reliably.

**Revisit with:** pooled years or wider age bands.

03 · DECEMBER RETENTION

### December wave

**Why deferred**
Limited days of observation were available before the 2026-01-01 cutoff.

**Revisit in:** Q1 2026, once 16 weeks elapse.

**Principle:** Prioritize analyses that are interpretable now; revisit deferred questions when the data can support them.

APPENDIX · SOURCES, LINEAGE & AI WORKFLOW

# Reproducible pipeline & human-in-the-loop AI workflow

All metrics, marts, and interactive visuals are generated reproducibly from immutable raw CSVs.

### 01 · CODE, NOTEBOOKS & PLOT ARCHIVE

**GitHub Repository** — vipps_da_interview_case

**Full Analysis Workbook** — notebooks/case_analysis.py

**Interactive Slide Deck** — notebooks/vip_presentation.py

**Static Plot Gallery** — plots/

### 02 · DATA LINEAGE & SEMANTIC MODEL

**Raw Case Data** `data/raw/` — transactions.csv, users.csv, merchants.csv, mcc.csv, experiment_exposure.csv, cards.csv (deferred)

**Derived Marts** `data/processed/` — df_user_spine.csv, df_exp_eval.csv, df_weekly_panel.csv

**Semantic Layer** `semantic/` — model.yml, marts.yml, erd.md

### 03 · TECH STACK & STATISTICAL METHODS

**Compute & QA** — Python, DuckDB, Pandas, NumPy, ruff, pytest

**Visualization** — Marimo + Altair (Vega-Lite)

**Statistical Inference** `scipy.stats` — Welch t-test, paired t-test, two-proportion z-test, complier-scaled ITT

### 04 · AI MODELS & HUMAN VERIFICATION

**Planning & Analytical Partner** — Gemini 4 Argon

**Coding & Implementation Model** — DeepSeek V4.1

**Human-in-the-Loop Governance** — every metric, cohort filter, and conclusion verified against raw data

**Thanks for your time & questions are welcome!**